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AERIAL VEHICLE AND METHOD OF CONTROLLING AERIAL VEHICLE

Abstract

An aerial vehicle includes a communication apparatus, a receiver that obtains a current position, a flight controller that controls flight and can transmit information on the current position with the communication apparatus, a plurality of drive apparatuses that drive respective different ones of a plurality of propulsive mechanisms, a distributor that distributes supplied electric power to each of the plurality of drive apparatuses, a power supply module that supplies electric power from a battery to the communication apparatus, the receiver, and the flight controller through a first power path and supplies electric power to the distributor through a second power path, and a disconnection apparatus that cuts off supply of electric power from the power supply module to the distributor through the second power path when abnormality occurs.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to an aerial vehicle and a method of controlling an aerial vehicle.

BACKGROUND ART

[0002] Various manned aircrafts and various unmanned aircrafts have conventionally been known as aerial vehicles.

[0003] For example, US Patent Publication No. 2017/0106986A1 (PTL 1) discloses an unmanned aircraft provided with a parachute. When the unmanned aircraft in PTL 1 senses a crash or the like, it expands the parachute and stops power feed to a motor that rotates a propeller. Specifically, in the unmanned aircraft in PTL 1, a power supply and the motor that rotates the propeller are directly connected to each other through a wire. When the unmanned aircraft in PTL 1 senses the crash or the like, it physically disconnects the power supply and the motor from each other. Power feed to the motor is thus stopped.

[0004] WO2019/216204A1 (PTL 2) discloses a current breaker included in an unmanned aircraft provided with a parachute. The current breaker cuts off supply of the current to an electric device of the unmanned aircraft. Occurrence of failure in expansion of the parachute at the time of (during) a crash of the unmanned aircraft is thus prevented. Furthermore, occurrence of accidental contact between an operating propeller and a human after the crash is prevented.

[0005] WO2017/030034A1 (PTL 3) discloses an unmanned aircraft in which a battery is chargeable by a power generator during flight for achieving a longer flight distance. When the unmanned aircraft senses a crash thereof, it stops charging from the power generator to the battery. Damage caused by an accident following the crash of the unmanned aircraft is thus prevented from spreading.

CITATION LIST

Patent Literature

[0006] PTL 1: US Patent Publication No. 2017/0106986A1

[0007] PTL 2: WO2019/216204A1

[0008] PTL 3: WO2017/030034A1

SUMMARY OF INVENTION

Technical Problem

[0009] The unmanned aircraft in PTL 1 includes an electronic speed controller (ESC) that drives the motor described above. The electronic speed controller is specifically a motor driver (amplifier) that controls the number of rotations of the motor by feed of a current to the motor. The electronic speed controller controls a flying speed of the unmanned aircraft. Therefore, even when the power supply and the motor are disconnected from each other as above, the current flows from the power supply to the motor through the electronic speed controller. Therefore, the unmanned aircraft in PTL 1 is actually unable to stop rotation of the propeller by disconnection as above.

[0010] The unmanned aircraft including the current breaker in PTL 2 causes the current breaker to

cut off supply of the current to the electric device thereof at the time of crash. Therefore, at the time of crash of the unmanned aircraft (descent with the use of the parachute), the unmanned aircraft is unable to calculate the current position and to communicate with an external device on the ground. Therefore, the external device on the ground is unable to specify the current position of the unmanned aircraft.

[0011] Even when the unmanned aircraft in PTL 3 senses the crash thereof, power is kept supplied from the battery to the motor that rotates the propeller. In other words, a propulsive mechanism (the propeller and the motor) of the unmanned aircraft keeps operating.

[0012] The present disclosure was made in view of the problems above, and an object thereof is to provide an aerial vehicle capable of deactivating all propulsive mechanisms by disconnection of a single power path and capable of transmitting information on a current position to an external device also after disconnection of the power path, and a method of controlling an aerial vehicle.

Solution to Problem

[0013] According to one aspect of the present disclosure, an aerial vehicle includes a battery, a plurality of propulsive mechanisms, a communication apparatus for wireless communication with an external device, a receiver that obtains a current position of the aerial vehicle by receiving radio waves emitted from a plurality of positioning satellites, a flight controller that controls flight of the aerial vehicle and can transmit information on the current position through the communication apparatus, a plurality of drive apparatuses each communicatively connected to the flight controller, the plurality of drive apparatuses driving respective different ones of the plurality of propulsive mechanisms, a distributor that distributes supplied electric power to each of the plurality of drive apparatuses, a power supply module that supplies electric power from the battery to the communication apparatus, the receiver, and the flight controller through a first power path and supplies electric power to the distributor through a second power path, and a disconnection apparatus that cuts off supply of electric power from the power supply module to the distributor through the second power path when predetermined abnormality occurs in the aerial vehicle.

[0014] According to another aspect of the present disclosure, an aerial vehicle supplies electric power from a contained battery through a first power path, to a flight controller that controls flight of the aerial vehicle, a communication apparatus for wireless communication with an external device, and a receiver that calculates a current position of the aerial vehicle, and supplies electric power through a second power path, to a distributor that distributes electric power to each of a plurality of drive apparatuses. The drive apparatuses drive respective different ones of a plurality of propulsive mechanisms based on a command from the flight controller. A method of controlling the aerial vehicle includes cutting off supply of electric power to the distributor through the second power path when predetermined abnormality occurs in the aerial vehicle and transmitting to the external device, information on the current position through the communication apparatus when the predetermined abnormality occurs.

Advantageous Effects of Invention

[0015] According to the present disclosure, when predetermined abnormality occurs in the aerial vehicle, all propulsive mechanisms can be deactivated by disconnection of a single power path, and information on the current position can be transmitted to an external device also after the power path is disconnected.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0016] FIG. 1 is a diagram showing a state that a drone is flying.

[0017] FIG. 2 is a diagram showing a state of descent of the drone with a parachute having been ejected.

[0018] FIG. 3 is a block diagram for illustrating a hardware configuration of the drone.

[0019] FIG. 4 is a diagram for illustrating a specific example of a disconnection apparatus.

[0020] FIG. 5 is a flowchart for illustrating a flow of processing performed in the drone.

[0021] FIG. 6 is a diagram showing a state before disconnection of a power path in a first modification of the disconnection apparatus.

[0022] FIG. 7 is a diagram showing a state after disconnection of the power path in the first modification of the disconnection apparatus.

[0023] FIG. 8 is a diagram for illustrating a second modification of the disconnection apparatus.

[0024] FIG. 9 is a block diagram for illustrating a hardware configuration of another drone.

DESCRIPTION OF EMBODIMENTS

[0025] An embodiment according to the present invention will be described below with reference to the drawings. In the description below, the same components and constituent elements have the same reference characters allotted and their labels and functions are also the same. Therefore, detailed description thereof will not be repeated.

[0026] An unmanned aircraft will be described below by way of example of an aerial vehicle. Specifically, a drone as the unmanned aircraft will be described by way of example. More specifically, a drone with rotary wings will be described by way of example. The present disclosure is also applicable to various manned aircrafts (other examples of the aerial vehicle) such as a “flying car.”

A. Overview of Drone

[0027] A drone according to the present embodiment is adapted to automatic control and manual control by a remote controller operation by an operator. FIG. 1 is a diagram showing a state that a drone is flying. FIG. 2 is a diagram showing a state of descent of the drone with a parachute having been ejected.

[0028] As shown in FIGS. 1 and 2, a drone 1 includes a main body 2, four arms 3A, 3B, 3C, and 3D, four propulsive mechanisms 4A, 4B, 4C, and 4D, an ejection apparatus 5, a communication unit 6, and a support member 18. Each of propulsive mechanisms 4A, 4B, 4C, and 4D includes a propeller 41 (see FIG. 2) and a motor 42.

[0029] For the sake of convenience of description, any one arm of arms 3A, 3B, 3C, and 3D will also be referred to as an “arm 3” below. Similarly, any one propulsive mechanism of propulsive mechanisms 4A, 4B, 4C, and 4D will also be referred to as a “propulsive mechanism 4.”

[0030] Arm 3 is attached to main body 2. Arm 3 has a base end and a tip end. Arm 3 has the base end attached to main body 2. In the present example, arm 3 extends from a side surface of main body 2.

[0031] Propulsive mechanism 4 is attached to the tip end of arm 3. Motor 42 is attached to the tip end of arm 3. Propeller 41 is attached to a rotation shaft (not shown) of motor 42.

[0032] Ejection apparatus 5 is attached to an upper surface (a top surface) of main body 2. A parachute 51 is accommodated in ejection apparatus 5. Ejection apparatus 5 ejects parachute 51 (see FIG. 2) when predetermined abnormality occurs in drone 1. Ejection apparatus 5 senses drop of drone 1 and instantaneously expands the parachute by activating a gas generator contained in ejection apparatus 5. An explosive gas generator (pyrotechnic) or a non-explosive gas generator such as a canister type gas generator may be employed as the gas generator.

[0033] Support member 18 extends upward from a portion of connection between arm 3 and main body 2 in the present example. Support member 18 is a hollow pipe. In the inside of support member 18, a wire for communication unit 6 passes. In the present example, communication unit 6 is supported by support member 18 as being located above main body 2.

[0034] Though drone 1 including four arms 3 and four propulsive mechanisms 4 is exemplified, the number of arms and the number of propulsive mechanisms are not limited as such.

B. Hardware Configuration and Performed Processing

[0035] FIG. 3 is a block diagram for illustrating a hardware configuration of drone 1. As shown in

FIG. 3, drone **1** can communicate with a terrestrial apparatus **900**. Terrestrial apparatus **900** includes a remote controller **901** and a terrestrial control station **902**. In FIG. 3, a main power path is shown with a solid line and a main signal path is shown with a dashed line.

[0036] In addition to propulsive mechanisms **4A**, **4B**, **4C**, and **4D**, ejection apparatus **5**, and communication unit **6** described above, drone **1** includes drive apparatuses (electronic speed controllers (ESCs)) **7A**, **7B**, **7C**, and **7D**, a flight controller (an autopilot) **8**, a battery **9**, a power supply module **10**, a disconnection apparatus (flight termination system (FTS)) **11**, a distributor **12**, a battery **13**, a trigger apparatus (auto trigger system (ATS)) **14**, and a sensor **15**. For the sake of convenience of description, any one drive apparatus of drive apparatuses **7A**, **7B**, **7C**, and **7D** will also be referred to as a “drive apparatus **7**” below.

[0037] As described above, propulsive mechanism **4** includes propeller **41** and motor **42**. Motor **42** includes a magnet **421** and a coil **422**. A type of motor **42** is not particularly limited. For example, a brushless direct-current (DC) motor, a brushed DC motor, or an alternating-current (AC) motor can be employed as motor **42**. Drive apparatus **7** includes a micro controller unit (MCU) **71** and a current driver **72**.

[0038] Flight controller **8** includes an MCU **81**, an inertial measurement unit (IMU) **82**, and a pressure altimeter **83**. MCU **81** includes an attitude estimator **811** and a determination unit **812** as functional blocks. MCU **81** is communicatively connected to inertial measurement unit **82** and pressure altimeter **83**.

[0039] Disconnection apparatus **11** includes a current switch **111**. Trigger apparatus **14** includes an MCU **141** and a sensor **142**. Sensor **142** detects an acceleration, an angular velocity, an attitude angle, and variation in altitude of drone **1**.

[0040] MCUs **71**, **81**, and **141** each specifically include a central processing unit (CPU), a memory such as a read only memory (ROM) and a random access memory (RAM), and an input/output (I/O) circuit which are not shown. A program and data are stored in the memory. For example, data indicating a flight route of drone **1** is stored in the memory of MUC **81**.

[0041] Communication unit **6** includes a global navigation satellite system (GNSS) receiver **61**, a wireless module **62**, and a long term evolution (LTE) module **63**.

[0042] Battery **9** is a storage battery (secondary battery). Battery **9** is connected to power supply module **10**.

[0043] Power supply module **10** supplies electric power from battery **9** to flight controller **8** and communication unit **6** through a power path **R1**. In the present example, power is fed to communication unit **6** through flight controller **8**. Power supply module **10** supplies electric power from battery **9** to distributor **12** through a power path **R2**. Specifically, power supply module **10** supplies electric power from battery **9** to distributor **12** through disconnection apparatus **11**. Power supply module **10** thus branches the power path from battery **9** into two power paths **R1** and **R2**.

[0044] In power path **R2**, a connector terminal of power supply module **10** and a connector terminal of distributor **12** are connected to each other through a wire. The connector terminal of power supply module **10** and the connector terminal of distributor **12** are preferably plated with gold in order to reduce heat generation due to contact resistance of the wire.

[0045] Distributor **12** distributes supplied electric power to each of four drive apparatuses **7**. Distributor **12** allows electric power supplied through power path **R2** to be supplied to each of four drive apparatuses **7**. Specifically, distributor **12** allows electric power to be supplied to current driver **72** of each of four drive apparatuses **7**. Though not shown with the solid line, electric power is supplied also to each MCU **71**. Current driver **72** and coil **422** of motor **42** are connected to each other through a wire.

[0046] Drive apparatus **7** is communicatively connected to flight controller **8**. Drive apparatus **7** drives corresponding propulsive mechanism **4** based on a command from flight controller **8**. Specifically, drive apparatus **7** adjusts propulsive force of corresponding propulsive mechanism **4** based on a command from flight controller **8**. Specifically, drive apparatus **7A** adjusts propulsive

force of corresponding propulsive mechanism **4A**. Similarly, drive apparatuses **7B**, **7C**, and **7D** adjust propulsive force of respective propulsive mechanisms **4B**, **4C**, and **4D**. Drive apparatus **7** thus functions as a control device that controls propulsive mechanism **4**.

[0047] Specifically, MUC **71** receives a command from MUC **81** of flight controller **8**. MCU **71** controls a current fed by current driver **72** to coil **422** of motor **42** based on the command. By such control of the current to be fed to coil **422**, a rotation speed of motor **42** is controlled. A speed of rotation of propeller **41** can thus be controlled.

[0048] GNSS receiver **61**, wireless module **62**, and LTE module **63** are communicatively connected to MCU **81** of flight controller **8**. GNSS receiver **61**, wireless module **62**, and LTE module **63** receive supply of electric power from flight controller **8**. GNSS receiver **61**, wireless module **62**, and LTE module **63** operate with electric power from battery **9**.

[0049] GNSS receiver **61** receives radio waves emitted from a plurality of positioning satellites (not shown) and obtains the current position of drone **1** based on received radio waves. GNSS receiver **61** periodically sends information indicating the current position to flight controller **8**. Specifically, GNSS receiver **61** periodically sends information on a latitude and a longitude of drone **1** as position information to MCU **81**.

[0050] Wireless module **62** is a communication module for communication with remote controller **901**. Communication between wireless module **62** and remote controller **901** is used in manual control of drone **1** within a range viewable by the operator. Therefore, this communication is real-time. A command from remote controller **901** is inputted to MCU **81** through wireless module **62**.

[0051] LTE module **63** is a communication module for communication with terrestrial control station **902** with a base station (not shown) being interposed. Communication between LTE module **63** and terrestrial control station **902** is used in flight of drone **1** over several kilometers by automatic control. In the present example, MUC **81** transmits position information of drone **1** to terrestrial control station **902** through LTE module **63**. Timing of transmission of the position information will be described later. Inertial measurement unit **82** periodically measures the angular velocity and the acceleration of drone **1** in a three-dimensional space. Pressure altimeter **83** periodically measures the altitude from the ground level and the sea level.

[0052] Flight controller **8** controls flight of drone **1**. Flight controller **8** realizes automatic control and manual control by a remote controller operation by the operator.

[0053] In automatic control, flight controller **8** controls flight of drone **1** based on a predetermined flight route. Specifically, MCU **81** of flight controller **8** implements autonomous flight of drone **1** based on the current position obtained by GNSS receiver **61**, a result of detection by inertial measurement unit **82**, and a result of detection (altitude) by pressure altimeter **83**. As described above, flight controller **8** can transmit information on the current position of drone **1** through LTE module **63** which is the communication apparatus.

[0054] More specifically, attitude estimator **811** of MCU **81** estimates an attitude of drone **1** based on a result of detection by inertial measurement unit **82**. Determination unit **812** of MCU **81** determines whether or not the estimated attitude is an abnormal attitude based on a predetermined reference. When MCU **81** determines the estimated attitude as the abnormal attitude, it transmits a predetermined signal to trigger apparatus **14**.

[0055] Flight controller **8** transmits the information on the current position of drone **1** to terrestrial control station **902** through LTE module **63**, based on occurrence of predetermined abnormality in drone **1**.

[0056] Without being limited as such, flight controller **8** may transmit the information on the current position of drone **1** to terrestrial control station **902** on condition that the drone has landed with the use of parachute **51**. When the current position does not change for a certain time period or longer, drone **1** can be determined as having landed. Alternatively, drone **1** can be determined as having landed based on the result of detection by pressure altimeter **83**. Drone **1** can be determined as having landed also based on a result of detection by inertial measurement unit **82**.

[0057] Ejection apparatus **5** contains the parachute. When the predetermined abnormality occurs in drone **1**, ejection apparatus **5** ejects parachute **51**. In the present example, ejection apparatus **5** ejects parachute **51** in any of automatic control and manual control.

[0058] When the predetermined abnormality occurs, disconnection apparatus **11** cuts off supply of electric power from power supply module **10** to distributor **12** through power path R2. Specifically, disconnection apparatus **11** cuts off supply of electric power to distributor **12** by an operation of current switch **111**. In the present example, disconnection apparatus **11** cuts off supply of electric power to distributor **12** in any of automatic control and manual control.

[0059] Trigger apparatus **14** is fed with power from battery **13**. Trigger apparatus **14** activates disconnection apparatus **11** and ejection apparatus **5** based on occurrence of predetermined abnormality in drone **1**. Specifically, MCU **141** sends a prescribed signal (command) to ejection apparatus **5**, so that ejection apparatus **5** ejects parachute **51** to the outside. As MCU **141** sends a prescribed signal (which will also be referred to as a “current break command signal” below) to disconnection apparatus **11**, current switch **111** of disconnection apparatus **11** is activated. As a result of the activation, supply of electric power from power supply module **10** to distributor **12** through power path R2 is cut off. Drone **1** is thus forced to stop autonomous flight.

[0060] The current break command signal above is desirably resistant to variation in voltage caused by electromagnetic noise. For example, the current break command signal is preferably a signal having as a variable, a time period for which the High level of a voltage is held.

[0061] Examples of the predetermined abnormality include the estimated attitude being an abnormal attitude and deviation from a scheduled flight route. Based on determination by MCU **81** of flight controller **8** that the attitude of drone **1** is abnormal or determination by MUC **81** as deviation from the flight route, MCU **141** of trigger apparatus **14** activates disconnection apparatus **11** and ejection apparatus **5**.

[0062] Without being limited as such, trigger apparatus **14** may determine whether or not predetermined abnormality has occurred based on a result of sensing by sensor **142**. A type of abnormality is not particularly limited so long as it is set in advance.

[0063] In the present example, when a prescribed condition is satisfied after trigger apparatus **14** causes disconnection apparatus **11** to cut off supply of electric power to distributor **12** through power path R2, trigger apparatus **14** causes ejection apparatus **5** to eject parachute **51**. Specifically, when a prescribed time period (for example, one second) has elapsed after trigger apparatus **14** caused disconnection apparatus **11** to cut off supply of electric power to distributor **12** through power path R2, trigger apparatus **14** causes ejection apparatus **5** to eject parachute **51**.

[0064] The prescribed time period is preferably within 1.5 second. The prescribed time period is preferably set to a time period longer than a time period in which rotation of propeller **41** stops.

[0065] Sensor **15** detects whether or not ejection apparatus **5** has been activated. Sensor **15** is communicatively connected to flight controller **8**. When the predetermined abnormality occurs, flight controller **8** transmits through LTE module **63** to terrestrial control station **902**, the fact of forced stop of the flight and the result of detection by sensor **15**. Flight controller **8** may transmit information indicating the type of abnormality to terrestrial control station **902**.

[0066] FIG. **4** is a diagram for illustrating a specific example of disconnection apparatus **11**. As shown in FIG. **4**, disconnection apparatus **11** includes a transistor **111A** that functions as current switch **111**.

[0067] Transistor **111A** is provided between power supply module **10** and distributor **12** and allows a flow of a current from power supply module **10** to distributor **12** through power path R2 while it is on. In the present example, transistor **111A** is a bipolar transistor. Specifically, transistor **111A** is an npn-type transistor.

[0068] When the predetermined abnormality described above occurs, disconnection apparatus **11** turns off transistor **111A**. As disconnection apparatus **11** switches a state of transistor **111A** from on to off, it cuts off supply of electric power from power supply module **10** to distributor **12** through

power path R2.

[0069] Transistor **111A** is not limited to the npn-type bipolar transistor. Transistor **111A** may be a pnp-type bipolar transistor. Alternatively, transistor **111A** may be a field effect transistor (unipolar transistor). In this case, transistor **111A** may be a junction field effect transistor (FET) or a MOS field effect transistor. Furthermore, transistor **111A** may be an insulated gate bipolar transistor (IGBT).

[0070] Transistor **111A** generates heat as a result of energization. In order to prevent a temperature of transistor **111A** from exceeding a rated temperature, disconnection apparatus **11** preferably includes a heat sink member that radiates heat of transistor **111A**. Disconnection apparatus **11** may include a substrate of an aluminum material for heat radiation or an electric fan for cooling instead of or together with the heat sink member.

[0071] When transistor **111A** is switched from on to off, a surge voltage is generated. In order to prevent break of transistor **111A** by the surge voltage, disconnection apparatus **11** preferably includes a snubber circuit. The snubber circuit includes at least one of a resistor and a capacitor.

C. Control Structure

[0072] FIG. **5** is a flowchart for illustrating a flow of processing performed in drone **1**. As shown in FIG. **5**, in step S1, drone **1** starts flying. In step S2, drone **1** determines whether or not the predetermined abnormality described above has occurred therein. When drone **1** determines that abnormality has not occurred (NO in step S2), the process proceeds to step S3. In step S3, drone **1** determines whether or not it has arrived at a destination. When drone **1** has determined that it has arrived at the destination (YES in step S3), it quits a series of processing. When drone **1** has determined that it has not arrived at the destination (NO in step S3), it has the process return to step S2.

[0073] When drone **1** determines that abnormality has occurred (YES in step S2), it activates disconnection apparatus **11** in step S4. In step S5, drone **1** determines whether or not a prescribed time period (for example, one second) has elapsed since activation of disconnection apparatus **11**.

[0074] When drone **1** determines that the prescribed time period has elapsed (YES in step S5), in step S6, it activates ejection apparatus **5**. In other words, drone **1** ejects parachute **51**. When drone **1** determines that the prescribed time period has not elapsed (NO in step S5), it has the process return to step S5.

[0075] In step S7, drone **1** periodically transmits the position information indicating the current position thereof to terrestrial apparatus **900**. Specifically, a base station apparatus (not shown) provided around drone **1** receives the position information of drone **1**, so that terrestrial control station **902** can obtain the position information.

D. Summary

[0076] (1) Referring to FIGS. **2** and **3**, as described above, drone **1** which is the unmanned aircraft includes (a) battery **9**, (b) a plurality of propulsive mechanisms **4** (**4A** to **4B**), (c) LTE module (communication apparatus) **62** for wireless communication with terrestrial apparatus **900** (specifically, terrestrial control station **902**) which is the external device, (d) GNSS receiver **61** that obtains the current position of drone **1** by receiving radio waves emitted from a plurality of positioning satellites, (e) flight controller **8** that controls flight of drone **1** and can transmit information on the current position through LTE module **62**, (f) a plurality of drive apparatuses **7** (**7A** to **7D**) each communicatively connected to flight controller **8** and driving respective different ones of the plurality of propulsive mechanisms **4**, (g) distributor **12** that distributes supplied electric power to each of the plurality of drive apparatuses **7**, (h) power supply module **10** that supplies electric power from battery **9** to LTE module **62**, GNSS receiver **61**, and flight controller **8** through power path R1 and supplies electric power to distributor **12** through power path R2, and (i) disconnection apparatus **11** that cuts off supply of electric power from power supply module **10** to distributor **12** through power path R2 when predetermined abnormality occurs in drone **1**.

[0077] According to the configuration above, when predetermined abnormality occurs, power path

R2 is disconnected. Therefore, supply of electric power from battery 9 through distributor 12 to all drive apparatuses 7 is cut off. Furthermore, as a result of cut-off of supply of electric power to drive apparatuses 7, drive of all propulsive mechanisms 4 is stopped. By thus disconnecting a single power path R2, all propulsive mechanisms 4 can be deactivated.

[0078] Even when power path R2 is disconnected, supply of electric power through power path R1 to LTE module 62, GNN receiver 61, and flight controller 8 continues. Therefore, according to drone 1, the information on the current position can be transmitted to terrestrial apparatus 900 also after power path R2 is disconnected. Drone 1 further includes ejection apparatus 5 that includes parachute 51 (FIG. 2) and ejects parachute 51 when the predetermined abnormality occurs. Therefore, even when all propulsive mechanisms 4 are deactivated by disconnection of power path R2 based on occurrence of the predetermined abnormality, crash of drone 1 can be prevented according to the configuration above.

[0079] Advantages of drone 1 will further be described below with reference to four comparative examples.

[0080] An example in which supply of electric power from the battery to the drone as a whole is stopped when the abnormality above occurs will be described as a first comparative example. In this case, at least the flight controller, the GNSS receiver, and a communication device for communication with a terrestrial facility do not operate. Therefore, in such a case, an operator on the ground is unable to obtain the latitude and the longitude of the drone. Therefore, the operator is unable to determine a position of descent of the drone. Accordingly, the operator is unable to collect the drone.

[0081] In contrast, according to drone 1 according to the present embodiment, even when the abnormality above occurs, flight controller 8, GNSS receiver 61, and LTE module 62 can keep operating, so that the operator can accurately know the position of descent of drone 1. The drone is thus readily collected.

[0082] An example in which the flight controller performs a propulsive mechanism deactivation function of the flight controller when abnormality occurs will be described as a second comparative example. In this case, when the flight controller itself fails (abnormality occurs therein), the plurality of propulsive mechanisms cannot be deactivated. Therefore, the plurality of propulsive mechanisms keep operating with the parachute having been expanded. In this case, the propulsive mechanisms may interfere with the parachute. For example, cords (lines) of the parachute may be entangled with the propellers. Therefore, the drone may crash. Even when the propulsive mechanisms do not interfere with the parachute, flight of the drone with the propulsive mechanisms continues.

[0083] In contrast, according to drone 1 according to the present embodiment, when the abnormality above occurs, the plurality of propulsive mechanisms 4 can be deactivated. Therefore, drone 1 can slowly descend with the use of parachute 51. In addition, flight of drone 1 with the use of the plurality of propulsive mechanisms 4 can be stopped.

[0084] An example in which the path for a signal sent from the flight controller to the drive apparatus for the propulsive mechanism is disconnected when abnormality occurs will be described as a third comparative example. In this case, when the drive apparatus fails (abnormality occurs therein), the propulsive mechanism cannot be deactivated.

[0085] According to drone 1 according to the present embodiment, on the other hand, even when drive apparatus 7 fails, supply of electric power to all drive apparatuses 7 is stopped as a result of disconnection of power path R2, and hence drive by all drive apparatuses 7 can be stopped. Therefore, according to drone 1, all propulsive mechanisms 4 can be deactivated.

[0086] A configuration in which a power supply circuit between the drive apparatus and the propulsive mechanism is disconnected when abnormality occurs will be described as a fourth comparative example. In this case, in order to deactivate all propulsive mechanisms, disconnection apparatuses as many as the propulsive mechanisms (that is, the drive apparatuses) are required.

Unless all disconnection apparatuses operate, at least one propulsive mechanism keeps operating. [0087] According to drone **1** according to the present embodiment, on the other hand, disconnection of a single power path **R2** suffices, and hence a single disconnection apparatus **11** suffices. Therefore, according to drone **1**, cost can be lower than in the fourth comparative example. A probability of failure of one disconnection apparatus **11** in the present embodiment over a certain period is compared with a probability of failure of any one of the plurality of disconnection apparatuses shown in the fourth comparative example. Then, in correspondence with a larger number of disconnection apparatuses, the latter probability can be concluded as being higher. Therefore, according to drone **1**, the probability of occurrence of such a situation that any propulsive mechanism keeps operating on the occurrence of abnormality can be lower than in the fourth comparative example where a plurality of disconnection apparatuses are required.

[0088] (2) Drone **1** includes trigger apparatus **14** that activates disconnection apparatus **11** and ejection apparatus **5** when the predetermined abnormality occurs. Trigger apparatus **14** causes ejection apparatus **5** to eject parachute **51** when a prescribed condition is satisfied after it caused disconnection apparatus **11** to cut off supply of electric power to distributor **12** through power path **R2**.

[0089] According to such a configuration, before ejection apparatus **5** is caused to eject parachute **51**, supply of electric power to all drive apparatuses **7** can be stopped.

[0090] Therefore, before ejection apparatus **5** is caused to eject parachute **51**, drive of all propulsive mechanisms **4** can be stopped. Therefore, according to drone **1**, as compared with the configuration in which parachute **51** is expanded while propulsive mechanism **4** is driven, the probability of entanglement of parachute **51** with propulsive mechanism **4** can be lowered.

[0091] (3) Flight controller **8** transmits information on the current position of drone **1** to terrestrial apparatus **900** through LTE module **62** based on occurrence of the predetermined abnormality. According to such a configuration, when abnormality occurs in drone **1**, terrestrial apparatus **900** can know the current position of drone **1**.

[0092] (4) Drone **1** includes sensor **15** that detects whether or not ejection apparatus **5** has been activated. When the predetermined abnormality occurs, flight controller **8** notifies terrestrial apparatus **900** of a result of detection by sensor **15**, through LTE module **62**. According to such a configuration, when abnormality occurs in drone **1**, terrestrial apparatus **900** can know whether or not drone **1** has expanded parachute **51**.

[0093] (5) Disconnection apparatus **11** includes transistor **111A** (FIG. **4**) provided between power supply module **10** and distributor **12**, transistor **111A** allowing a flow of the current from power supply module **10** to distributor **12** through power path **R2** while it is on. When the predetermined abnormality occurs, drone **1** turns off transistor **111A**. According to such a configuration, supply of electric power from power supply module **10** to distributor **12** can be stopped. Therefore, drive of all propulsive mechanisms **4** can be stopped.

[0094] (6) Each propulsive mechanism **4** includes propeller **41** and motor **42** that rotates propeller **41**. Each drive apparatus **7** drives motor **42**.

E. Modification

[0095] (1) A first modification of disconnection apparatus **11** will be described. FIG. **6** is a diagram showing a state before disconnection of power path **R2** in the first modification of the disconnection apparatus. FIG. **7** is a diagram showing a state after disconnection of power path **R2** in the first modification of the disconnection apparatus.

[0096] As shown in FIG. **6**, drone **1** includes a disconnection apparatus **11A** instead of disconnection apparatus **11**. Disconnection apparatus **11A** includes an igniter **150**, a cut-off chamber **160** containing an internal space, and a rupture plate **163** that is damaged and cleaved as a result of application of heat and a pressure produced by activation of igniter **150** and disconnects power path **R2**. When the abnormality above occurs in drone **1**, trigger apparatus **14** transmits a prescribed signal (command) to disconnection apparatus **11A** so that igniter **150** is driven.

[0097] Igniter **150** ignites a flame, and includes an ignition portion **151** containing an ignition agent (not shown) that ignites a flame by being fired and burnt at the time of activation and a resistor (not shown) for ignition of the ignition agent and a pair of terminal pins **152** connected to ignition portion **151**.

[0098] When abnormality of drone **1** is sensed, a prescribed amount of current flows to the resistor through the pair of terminal pins **152** upon reception of the command from trigger apparatus **14**. As the current flows to the resistor, Joule heat is generated in the resistor and the ignition agent starts burning. Flames at a high temperature produced by burning rupture a squib cup (not shown) where the ignition agent is accommodated.

[0099] A through hole **161** is provided in a peripheral wall of cut-off chamber **160** and a through hole **162** is provided in another portion of the peripheral wall. Power path R2 lies like a bridge between through holes **161** and **162**. Power path R2 is implemented by a bus bar. Power path R2 is formed, for example, from a metal plate or a metal wire. Power path R2 has one end connected to power supply module **10** and has the other end connected to distributor **12**.

[0100] When abnormality is sensed in drone **1** and when the prescribed amount of current is supplied to the pair of terminal pins **152** of igniter **150** in response to the command from trigger apparatus **14**, rupture plate **163** is damaged by heat and the pressure produced by activation of ignition portion **151** as shown in FIG. 7. In this case, rupture plate **163** is damaged in such a manner that a central portion thereof is cleaved and bent toward power path R2. Power path R2 (specifically, the bus bar) is cut off by damaged rupture plate **163**.

[0101] As a result of such a process, supply of electric power from battery **9** through distributor **12** to all drive apparatuses **7** can be stopped. Therefore, drive of all propulsive mechanisms **4** can be stopped.

[0102] Instead of cutting the bus bar with rupture plate **163** as above, an actuator may cut the bus bar. For example, the bus bar may be cut by movement of a piston by a gas pressure. Alternatively, instead of such cutting, a fuse may be provided on an electric circuit and a current greater than a rated current may be fed to the fuse to blow the fuse. According to such a configuration as well, drive of all propulsive mechanisms **4** can be stopped similarly to the above. [0103] (2) FIG. 8 is a diagram for illustrating a second modification of disconnection apparatus **11**. As shown in FIG. 8, drone **1** includes a disconnection apparatus **11B** instead of disconnection apparatus **11**.

Disconnection apparatus **11B** includes a relay switch **111B**. Relay switch **111B** includes a coil **171**, a moving contact **172**, and a fixed contact **173**.

[0104] In relay switch **111B**, a current is fed to coil **171** so that moving contact **172** comes in contact with fixed contact **173**. In other words, relay switch **111B** is turned on. When feed of the current to coil **171** is stopped, moving contact **172** moves away from fixed contact **173** and relay switch **111B** is turned off.

[0105] While the abnormality described above is not occurring in such a configuration, the current is kept fed to coil **171**. When the abnormality occurs, feed of the current to coil **171** is stopped. According to such control, an effect as in use of transistor **111A** is obtained. Instead of relay switch **111B**, a contactor (an electromagnetic contactor) may be employed. [0106] (3) Battery **9** is preferably attachable and removable. When battery **9** is running low, battery **9** is removed and a completely charged battery is attached, so that drone **1** can fly without waiting for charging. Drone **1** can thus efficiently operate. [0107] (4) Though the configuration in which communication unit **6** receives supply of electric power from battery **9** through power path R1 is described above by way of example, limitation thereto is not intended.

[0108] FIG. 9 is a block diagram for illustrating a hardware configuration of a drone **1A** which is a modification of drone **1**. As shown in FIG. 9, in the present modification, communication unit **6** receives supply of electric power by battery **9** through a power path R3 from power supply module **10**, instead of reception of supply of electric power by battery **9** from flight controller **8**.

[0109] Thus, in the present modification, power supply module **10** supplies electric power from

battery **9** to flight controller **8** through power path **R1**, to distributor **12** through power path **R2**, and to GNSS receiver **61**, wireless module **62**, and LTE module **63** through power path **R3**. According to such a configuration as well, an effect similar to that of drone **1** is obtained by cut-off by disconnection apparatus **11**, of supply of electric power from power supply module **10** to distributor **12** through power path **R2** on the occurrence of predetermined abnormality. [0110] (5) Though the configuration in which electric power is supplied from battery **9** to communication unit **6** is described above by way of example, limitation thereto is not intended. Drone **1** and drone **1A** may further include another not-shown battery, and another battery may be configured to supply electric power to communication unit **6** (specifically, GNSS receiver **61**, wireless module **62**, and LTE module **63**).

[0111] [Additional Aspect] An aerial vehicle, in which the battery includes a first battery (corresponding to battery **9**) and a second battery (corresponding to another battery), power supply module **10** supplies electric power from the first battery to the flight controller through a first power path and to the distributor through a second power path and supplies electric power from the second battery to the communication apparatus and the receiver through a third power path.

[0112] The embodiments disclosed herein are illustrative and not restricted only to contents above. The scope of the present invention is defined by the terms of the claims and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

REFERENCE SIGNS LIST

[0113] **1**, **1A** drone; **2** main body; **3** arm; **3A**, **3B**, **3C**, **3D** arm; **4A**, **4B**, **4C**, **4D** propulsive mechanism; **5** ejection apparatus; **6** communication unit; **7A**, **7B**, **7C**, **7D** drive apparatus; **8** flight controller; **9**, **13** battery; **10** power supply module; **11**, **11A**, **11B** disconnection apparatus; **12** distributor; **14** trigger apparatus; **15**, **142** sensor; **18** support member; **41** propeller; **42** motor; **51** parachute; **61** GNSS receiver; **62** wireless module; **63** LTE module; **71**, **81**, **141** MCU; **72** current driver; **82** inertial measurement unit; **83** pressure altimeter; **111** current switch; **111A** transistor; **111B** relay switch; **150** igniter; **151** ignition portion; **152** terminal pin; **160** cut-off chamber; **161**, **162** through hole; **163** rupture plate; **171**, **422** coil; **172** moving contact; **173** fixed contact; **421** magnet; **811** attitude estimator; **812** determination unit; **900** terrestrial apparatus; **901** remote controller; **902** terrestrial control station; **R1**, **R2** power path.

Claims

1. An aerial vehicle comprising: a battery; a plurality of propulsive mechanisms; a communication apparatus for wireless communication with an external device; a receiver that obtains a current position of the aerial vehicle by receiving radio waves emitted from a plurality of positioning satellites; a flight controller that controls flight of the aerial vehicle and can transmit information on the current position through the communication apparatus; a plurality of drive apparatuses each communicatively connected to the flight controller, the plurality of drive apparatuses driving respective different ones of the plurality of propulsive mechanisms; a distributor that distributes supplied electric power to each of the plurality of drive apparatuses; a power supply module that supplies electric power from the battery to the communication apparatus, the receiver, and the flight controller through a first power path and supplies electric power to the distributor through a second power path; and a disconnection apparatus that cuts off supply of electric power from the power supply module to the distributor through the second power path when predetermined abnormality occurs in the aerial vehicle.
2. The aerial vehicle according to claim 1, further comprising an ejection apparatus that includes a parachute and ejects the parachute when the predetermined abnormality occurs.
3. The aerial vehicle according to claim 2, further comprising a trigger apparatus that activates the disconnection apparatus and the ejection apparatus when the predetermined abnormality occurs, wherein the trigger apparatus causes the ejection apparatus to eject the parachute when a prescribed

condition is satisfied after the trigger apparatus causes the disconnection apparatus to cut off supply of electric power to the distributor through the second power path.

4. The aerial vehicle according to claim 1, wherein the flight controller transmits information on the current position to the external device through the communication apparatus, based on occurrence of the predetermined abnormality.

5. The aerial vehicle according to claim 2, further comprising a sensor that detects whether the ejection apparatus has been activated, wherein when the predetermined abnormality occurs, the flight controller notifies the external device of a result of detection by the sensor, through the communication apparatus.

6. The aerial vehicle according to claim 1, wherein the disconnection apparatus includes a transistor provided between the power supply module and the distributor, the transistor allowing a flow of the current from the power supply module to the distributor through the second power path while the transistor is on, and when the predetermined abnormality occurs, the disconnection apparatus turns off the transistor.

7. The aerial vehicle according to claim 1, wherein the aerial vehicle is an unmanned aircraft, each of the propulsive mechanisms includes a propeller and a motor that rotates the propeller, and each of the drive apparatuses drives the motor.

8. A method of controlling an aerial vehicle, the aerial vehicle supplying electric power from a contained battery through a first power path, to a flight controller that controls flight of the aerial vehicle, a communication apparatus for wireless communication with an external device, and a receiver that calculates a current position of the aerial vehicle, and supplying electric power through a second power path, to a distributor that distributes electric power to each of a plurality of drive apparatuses, the drive apparatuses driving respective different ones of a plurality of propulsive mechanisms based on a command from the flight controller, the method comprising: cutting off supply of electric power to the distributor through the second power path when predetermined abnormality occurs in the aerial vehicle; and transmitting to the external device, information on the current position through the communication apparatus when the predetermined abnormality occurs.

9. The aerial vehicle according to claim 2, wherein the flight controller transmits information on the current position to the external device through the communication apparatus, based on occurrence of the predetermined abnormality.

10. The aerial vehicle according to claim 3, wherein the flight controller transmits information on the current position to the external device through the communication apparatus, based on occurrence of the predetermined abnormality.

11. The aerial vehicle according to claim 3, further comprising a sensor that detects whether the ejection apparatus has been activated, wherein when the predetermined abnormality occurs, the flight controller notifies the external device of a result of detection by the sensor, through the communication apparatus.

12. The aerial vehicle according to claim 2, wherein the disconnection apparatus includes a transistor provided between the power supply module and the distributor, the transistor allowing a flow of the current from the power supply module to the distributor through the second power path while the transistor is on, and when the predetermined abnormality occurs, the disconnection apparatus turns off the transistor.

13. The aerial vehicle according to claim 3, wherein the disconnection apparatus includes a transistor provided between the power supply module and the distributor, the transistor allowing a flow of the current from the power supply module to the distributor through the second power path while the transistor is on, and when the predetermined abnormality occurs, the disconnection apparatus turns off the transistor.

14. The aerial vehicle according to claim 2, wherein the aerial vehicle is an unmanned aircraft, each of the propulsive mechanisms includes a propeller and a motor that rotates the propeller, and each

of the drive apparatuses drives the motor.

15. The serial vehicle according to claim 3, wherein the aerial vehicle is an unmanned aircraft, each of the propulsive mechanisms includes a propeller and a motor that rotates the propeller, and each of the drive apparatuses drives the motor.
